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Department of Information technology

Topic: Automated Collage Timetable Generator

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OBJECTIVE

- Develop a Java-based system to automate timetable generation.
- Implement both web and desktop interfaces using a shared database.
- Ensure conflict-free schedules with respect to time, faculty, and rooms.

ABSTRACT

The Automated College Timetable Generator is a Java-based system designed to eliminate the complexities of manual timetable creation. It generates conflict-free class schedules based on faculty availability, classroom resources, and course requirements. The system is implemented both as a web and desktop application sharing a common database and supports exporting the final timetable to an Excel file for manual adjustments.

SCOPE

This system will cover generation of class schedules for multiple departments, considering faculty, subjects, and classroom availability. It will provide both a desktop and web interface, use a shared MySQL database, and allow Excel export. The logic can be extended to include electives, lab sessions, or integration with mobile apps.

The system can be expanded to include mobile access, integration with academic calendars, notifications for schedule changes, and advanced AI-based optimization for better scheduling.

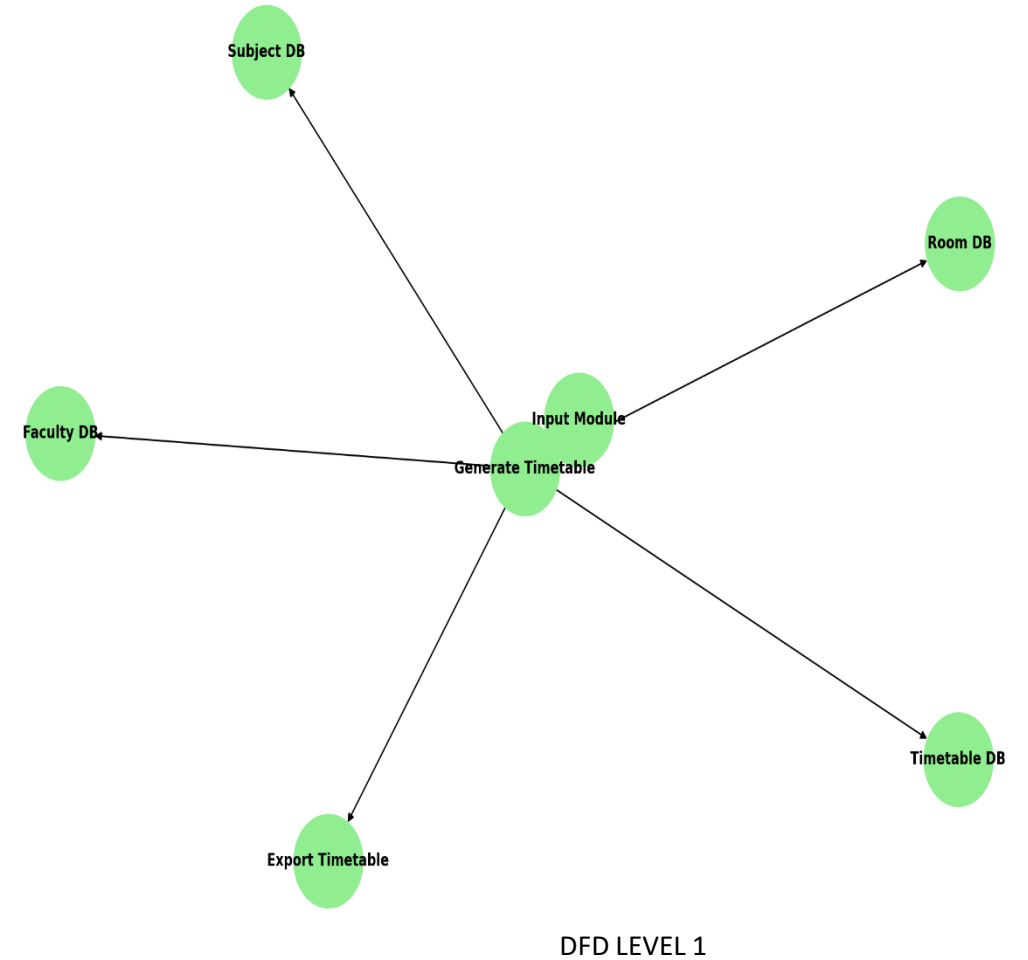
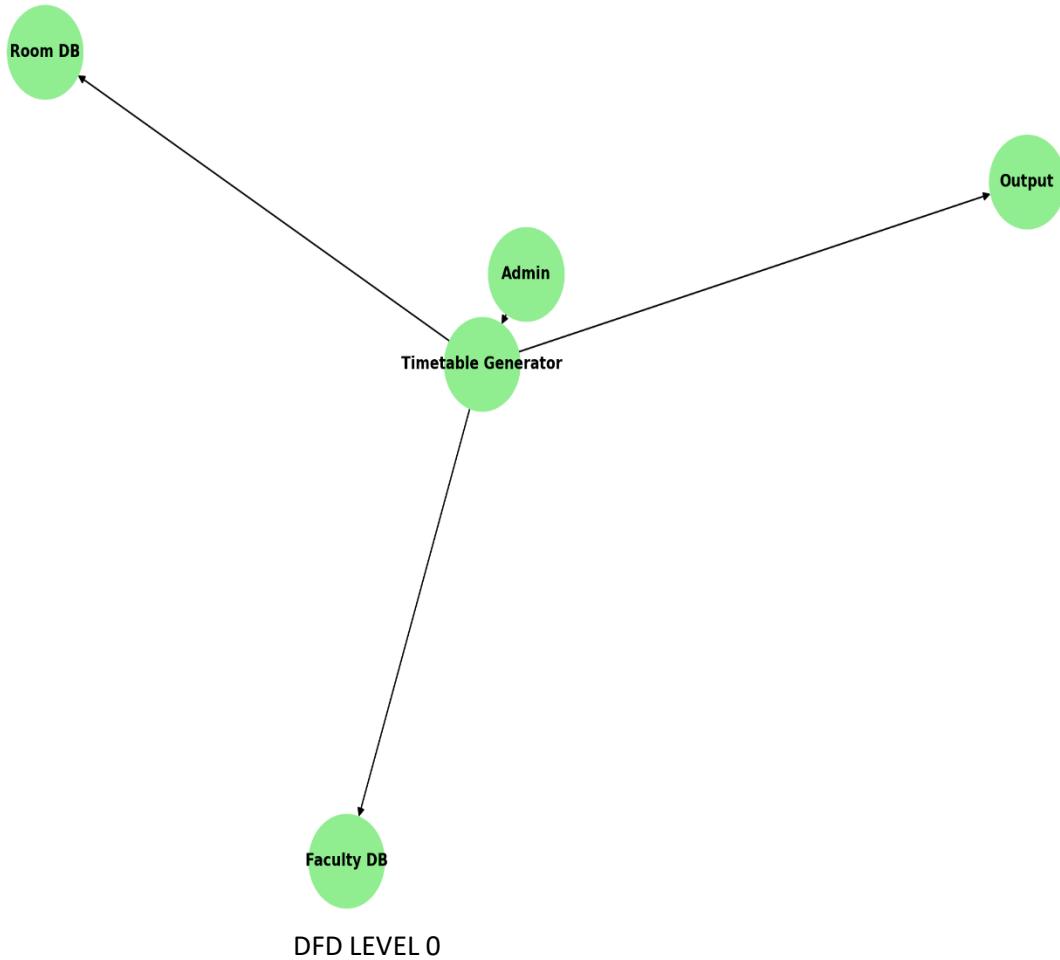
PROPOSED APPROACH

- Build a modular architecture for timetable generation.
- Store data in a centralized MySQL database shared by both applications.
- Implement an algorithm that avoids faculty and room conflicts.
- Allow timetable viewing, editing, and exporting.
- Maintain performance and usability across platforms.

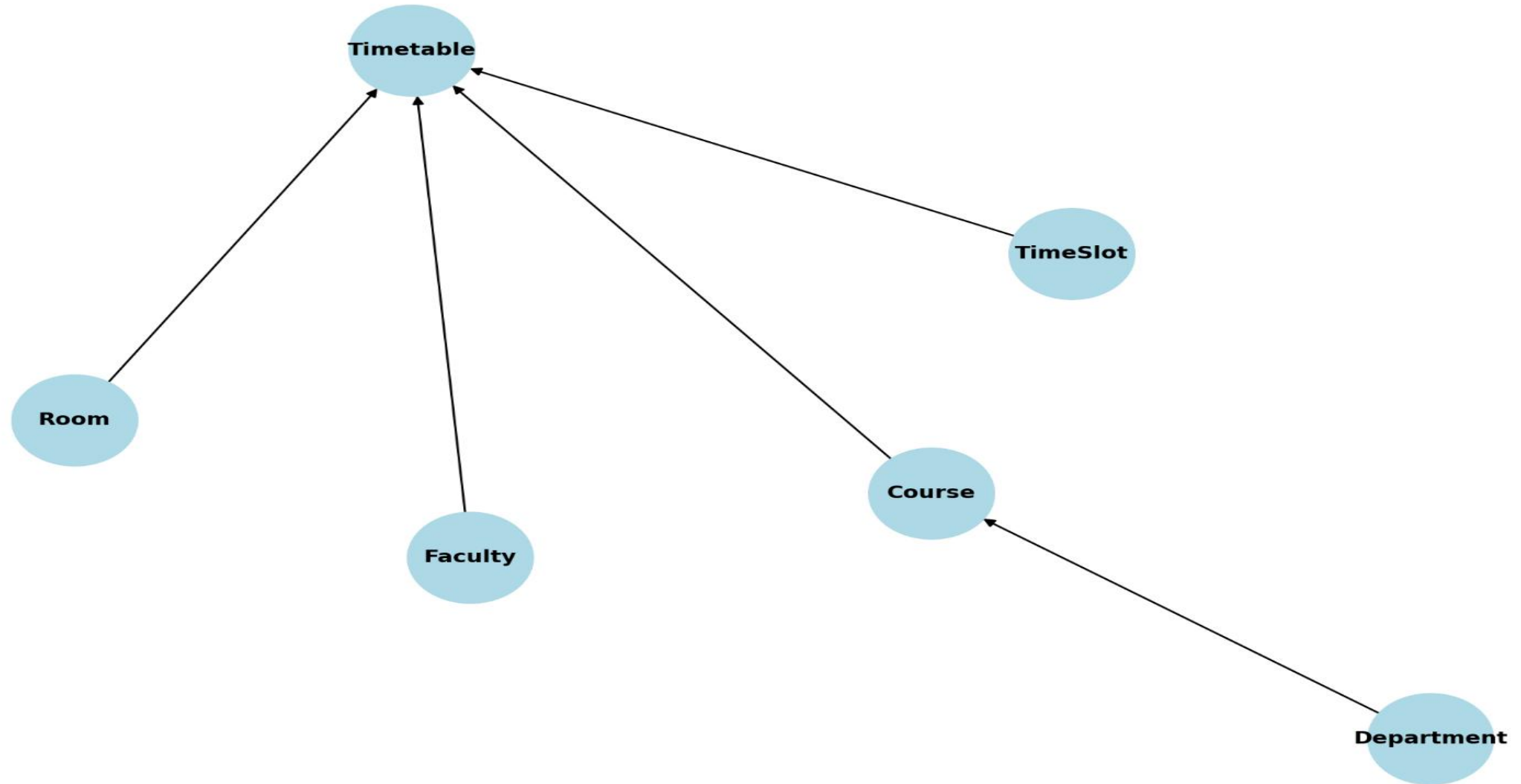
SYSTEM ARCHITECTURE

- **Admin Panel:** Manage departments, subjects, and faculty details.
- **Faculty Module:** View and request schedule changes.
- **Timetable Generator:** Core logic applying constraints to assign classes.
- **Database:** Central MySQL storing all records.
- **Output:** Export timetable to Excel for final review.

• DFD DIAGRAMS



• ER DIAGRAM



USED TECH STACK

- **Frontend (Desktop):** Java Swing / JavaFX
- **Frontend (Web):** JSP / HTML / CSS / JavaScript
- **Backend:** Java (Servlets / Spring Boot)
- **Database:** MySQL
- **Tools:** NetBeans / Eclipse / MySQL Workbench

OVERVIEW OF USER INTERFACE

- **Admin Interface:** Add, edit, or delete subjects, rooms, and faculty.
- **Faculty Interface:** View allocated schedules.
- **Web UI:** Responsive, simple HTML-based layout.
- **Desktop UI:** JavaFX forms for local administration.

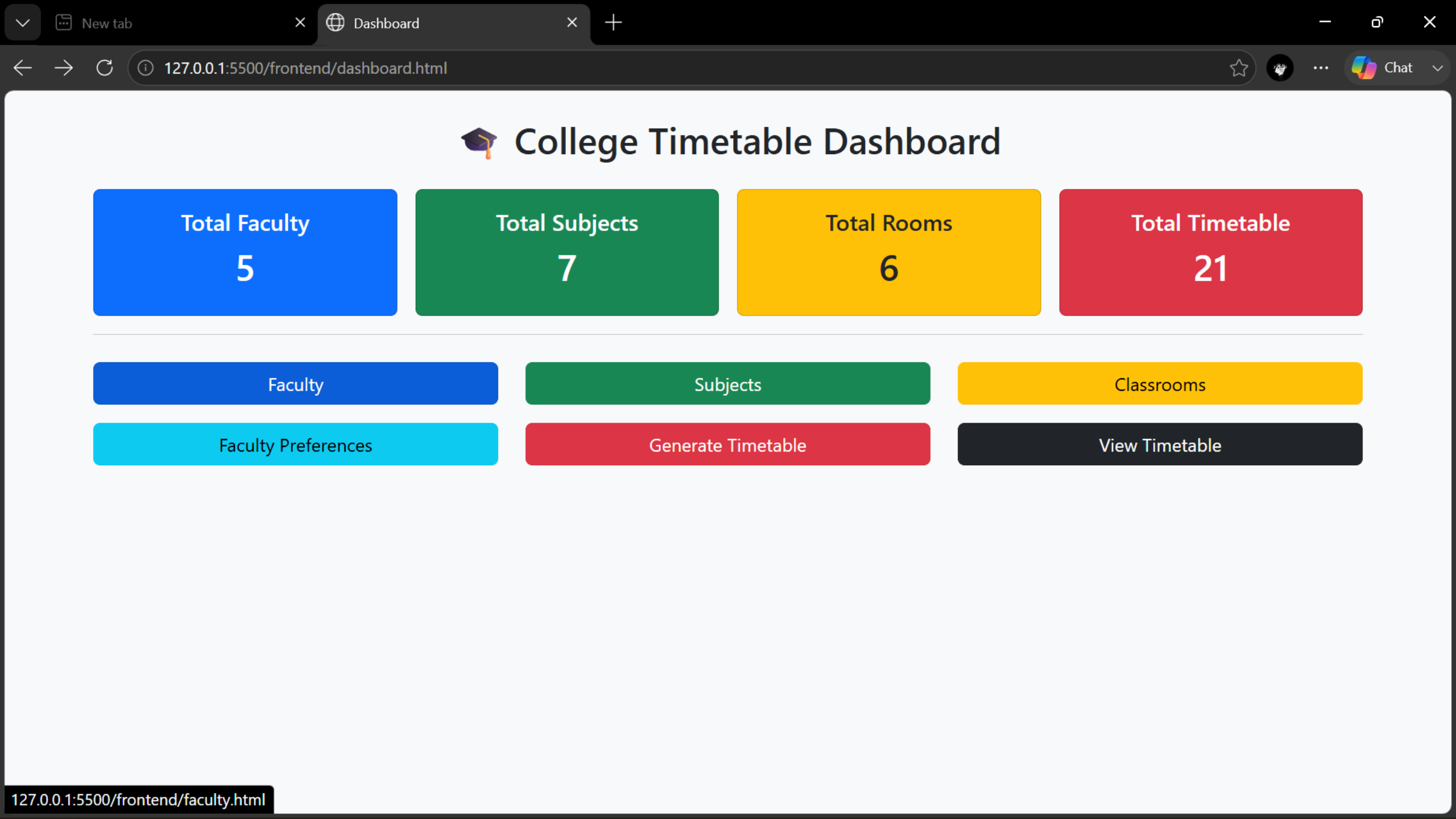
Admin login

Email / Username

Password

Login

[Forgot Password?](#) [Register](#)



College Timetable Dashboard

Total Faculty

5

Total Subjects

7

Total Rooms

6

Total Timetable

21

Faculty

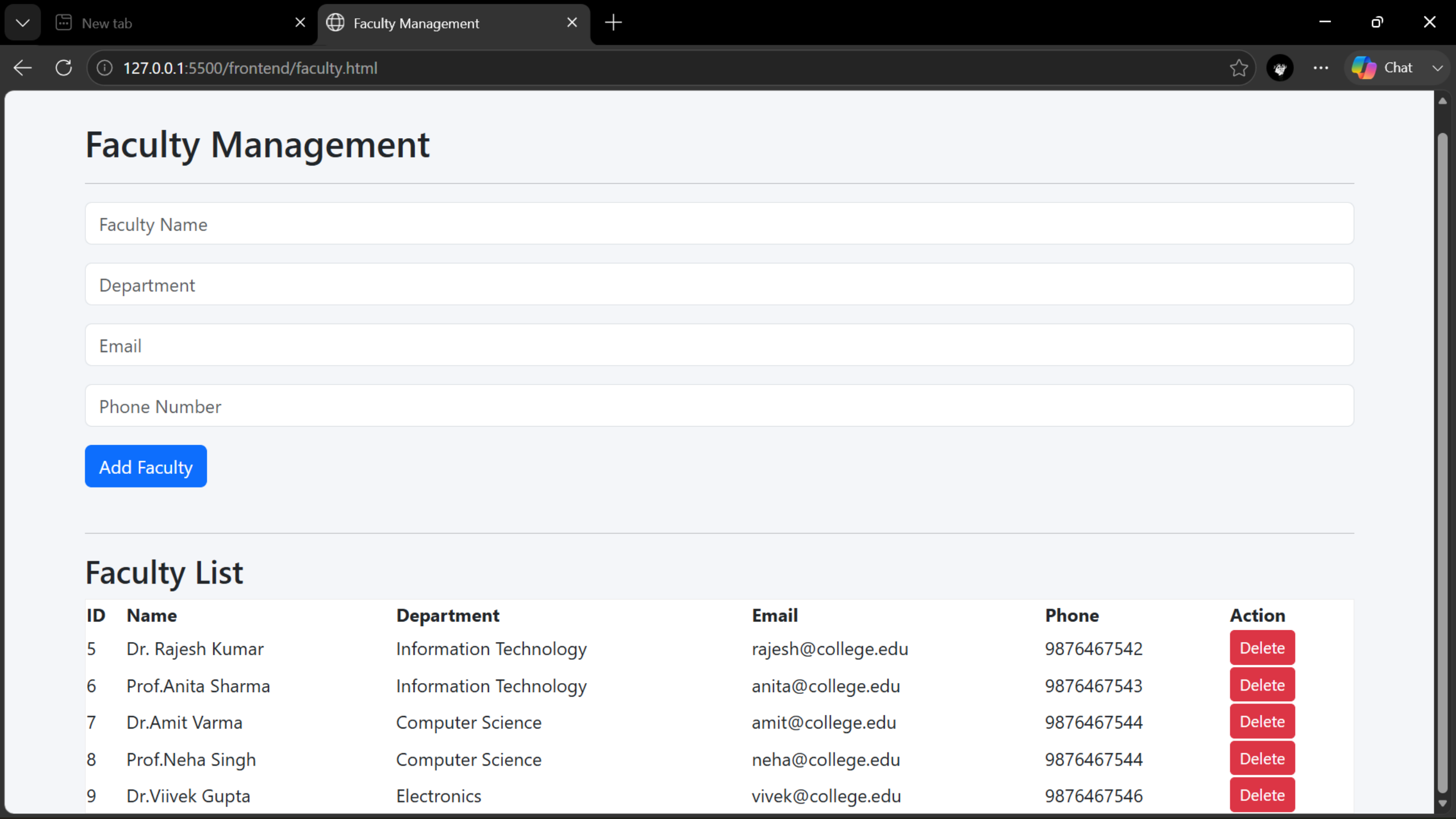
Subjects

Classrooms

Faculty Preferences

Generate Timetable

View Timetable



New tab

Subjects Management

127.0.0.1:5500/frontend/subjects.html

Star

GitHub

Chat

Subjects Management

Subject Name

Semester

Department

Faculty

Faculty

Select Faculty

Select Faculty

Theory Hours

Practical Hours

Add Subject

The image shows a VS Code workspace for a web application. The Explorer sidebar on the left lists the project structure, including folders for backend, database, and frontend, and files like subjects.html and timetable.html. The main editor displays the content of subjects.html, which is an HTML file for a 'Subjects Management' page. The code includes a DOCTYPE declaration, meta tags for charset and viewport, a title, and links to Bootstrap CSS and a custom stylesheet. The body contains a container with a heading, a horizontal rule, and a form with a text input field. The bottom status bar shows the server is running on http://localhost:3000 and MySQL is connected successfully. On the right, the Chat sidebar is open, displaying a message about asking about code and a code snippet for styling the body and card elements.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Subjects Management</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="container mt-5">
    <h1>Subjects Management</h1>
    <hr>
    <form id="subjectForm">
      <div class="mb-3">
        <input type="text" id="subject_name" class="form-control" placeholder="Subject Name">
      </div>
    </form>
  </div>
</body>
```

Server is running on http://localhost:3000
MySQL Connected Successfully

body{
background:#f4f6f9;
}
.card{
border-radius:15px;
}

Classroom Management

Room Name

Theory

Capacity

Add Classroom

Classroom List

ID	Room Name	Room Type	Capacity	Action
7	A101	Theory	60	Delete
8	A102	Theory	60	Delete
9	A201	Theory	80	Delete
10	Lab-1	Lab	30	Delete

The image shows a VS Code workspace for a web application. The Explorer sidebar on the left displays the project structure, including a 'COLLEGE-TIM...' folder with subfolders for 'backend', 'database', and 'frontend'. The 'frontend' folder contains 'css', 'images', 'js', and 'classrooms.html'. The main editor window shows the code for 'classrooms.html', which includes HTML structure for a form with a text input and a select dropdown. The bottom status bar indicates the server is running on http://localhost:3000 and MySQL is connected successfully. The right sidebar shows the CHAT interface with a prompt 'Ask about your code' and a code snippet for styling the body and .card elements.

OVERVIEW OF BACKEND SURFACE

- Database connectivity using MY SQL Workbench.
- Timetable generation logic implemented using constraint-based scheduling.
- Data validation and duplication checks.
- PDF format download.

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File

Edit

Selection

View

Go

Run

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College-Timetable

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Sign In

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✖

EXPLORER

⋮

JS server.js

login.html

JS login.js

index.html

dashboard.html

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CHAT

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✖

OPEN EDITORS2 unsaved

JS const express = require("..."

JS server.js backend

login.html frontend

JS login.js frontend\js

index.html frontend

dashboard.html frontend

faculty.html frontend

subjects.html frontend

classrooms.html frontend

COLLEGE-TIMETABLE

backend

node_modules

package-lock.json

package.json

JS server.js

database

frontend

backend > JS server.js > db

1 console.log("***** NEW CODE RUNNING *****");

2 const express = require("express");

3 const cors = require("cors");

4 const mysql = require("mysql2");

5

6 const app = express();

7

8 app.use(cors());

9 app.use(express.json());

10

11 // MySQL Connection

12 const db = mysql.createConnection({

13 host: "localhost",

14 user: "root",

15 password: "Rajdeep@123",

16 database: "college_timetable"

17 });

18

19 db.connect((err) => {

20 if (err) {

21 console.log("Database Connection Failed");

22 console.log(err);

23 } else {

24 console.log("MySQL Connected Successfully");

25 }

26 });

27

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

node

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Server is running on http://localhost:3000

MySQL Connected Successfully

OUTLINE

TIMELINE

Ask about your code

AI responses may be inaccurate

[Generate Agent Instructions](#) to onboard AI onto your codebase.

JS server.js

body{

background:#f4f6f9;

}

.card{

border-radius:15px;

}

+ | ? | Models

Local

Ln 15, Col 32

Spaces: 4

UTF-8

CRLF

{ }

JavaScript

🔍

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📄

📄

📄

Port : 5500

🔔

[illegible]

New tab

College Timetable Generator

127.0.0.1:5500/frontend/timetable.html

☆🐼⋮Chat

College Timetable Generator

DashboardFacultySubjectsRoomsPreferencesTimetable

Weekly College Timetable

Generate Timetable

Download PDF

Time	Monday	Tuesday	Wednesday	Thursday	Friday
09:00 AM - 10:00 AM	Computer Network Lab-1	Software Engineering A201	Java Programming Lab-2		
10:00 AM - 11:00 AM	Digital Electronics Lab-2	Software Engineering Lab-3	Java Programming Lab-1		
11:00 AM - 12:00 PM	Digital Electronics A102	Software Engineering Lab-2			
12:00 PM - 01:00 PM					
02:00 PM - 03:00 PM					
03:00 PM - 04:00 PM					

FUTURE WORK

- Integration with mobile app for timetable notifications.
- AI-based optimization for improved resource allocation.
- Automatic conflict detection during faculty updates.
- Integration with academic calendar and holidays.

CONCLUSION

The Automated College Timetable Generator streamlines the process of academic scheduling by eliminating human errors and optimizing resources. Its hybrid design—both desktop and web-based—offers flexibility, scalability, and ease of use. With future enhancements, this system can evolve into a comprehensive academic management tool supporting smart scheduling and institutional efficiency.

THANK YOU